

Re: [BRDF] Debugging initial measurements

From Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Date Fri 6/26/2026 11:49 PM
To Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>

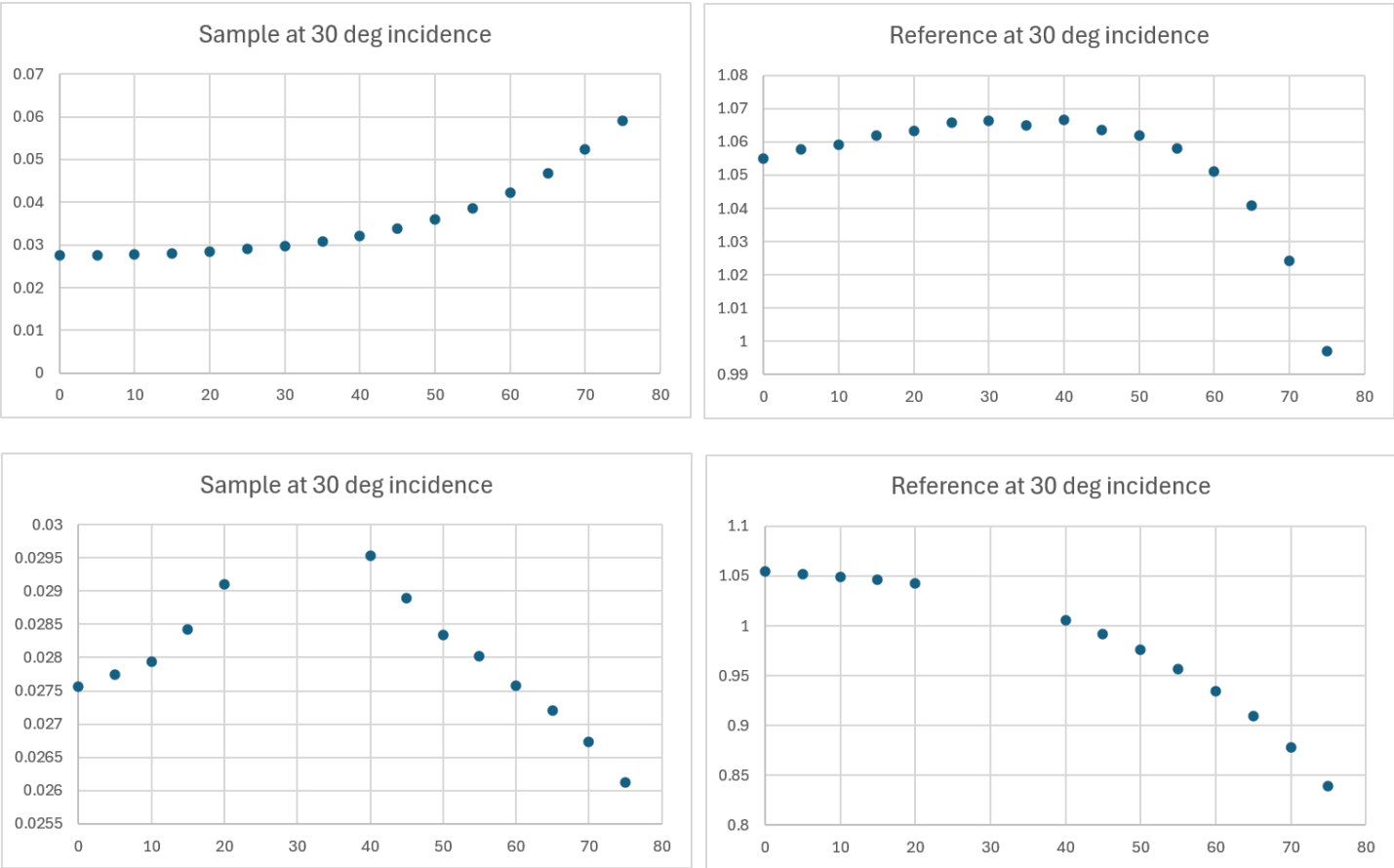
Dear Nik,

Thank you very much for your detailed and thoughtful explanation. I truly appreciate it.
I agree that the unique trend observed on the opposite side is likely related to the material properties. In particular, black paint surfaces (or any randomly scattering, non-Lambertian materials) tend to exhibit more specular-like reflection at larger incidence angles. At shallow incidence angles, the beam effectively interacts with smaller-scale surface roughness, which can lead to increased specular reflection.

The back reflection, however, is somewhat surprising. In general, black coating treatments are not expected to have microstructures that would induce a backscattering peak. One possible explanation is that the proximity of the light source and detector allows for a partial direct beam path from source to detector, effectively bypassing reflection from the sample.
That said, the signal peak value is approximately 0.003, which is quite small and likely not a major concern for stray light control.

Best regards,
Heejoo

From: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>
Sent: Friday, June 26, 2026 11:38 AM
To: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>
Subject: RE: [BRDF] Debugging initial measurements



Just one very quick example of what I mean when I mention that the angular trends from your sample are not seen for our reference.
(And thus I believe may be real.)

As you would imagine the bottom plots with the skipped data points are with the radiometer on the same side as the source (data points skipped for radiometer shading the sample). Top plots show radiometer on the opposite side of the source.

So, it appears that your sample does have a bit of a backscatter peak, however small. It does not seem to have much of a specular peak (not even as much as the slight hump for the Spectralon reference!), but does seem to scatter light more at those glancing angles.

From: Anderson, Nikolaus J - (nja)

Sent: Friday, June 26, 2026 10:41 AM

To: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>

Cc: Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>

Subject: RE: [BRDF] Debugging initial measurements

Heejoo,

Jacob and I spoke yesterday about these initial measurements. We have some work to do to ensure any comparisons to your scatterometer make sense, and understand details for how our lab works for low reflectance samples. Some answers to your questions and other things that came up:

- We realize that we'll have to agree on a consistent nomenclature for displaying and logging angles. In our nomenclature, zenith angles are always positive, and azimuth angles can be $0 - 360^\circ$ (or $-180 - 180^\circ$, although we tend to stick to all positive values). Certainly, this doesn't help make simple plots like Jacob was doing, so I'm sure we can collaborate on agreed upon ways to present data.
- Our VNIR radiometer has about a 1° FFOV and a 25mm aperture, at a typical viewing distance of ~820mm, so it sees a spot on the sample roughly 40mm on the sample.
- Our SWIR radiometer has about a 5.5° FFOV but a smaller entrance pupil so it should see a roughly 90mm spot on the sample.
- We have equipment to reduce the spot size of the FOV on the sample for both radiometers. So far, we have decided not to use them based on the larger size of your sample (which should yield more realistic results). We will want to use them if we end up trying smaller samples for you.
- After some further research and better recollection, I think I can confirm some of the corrections Jacob made to better compare with the scatterometer (BRDF) were unnecessary. Please see the section below which I copied from a previous student (Josie Maxwell), who worked on our current BRDF processing code. Based on this and other documentation, I believe using these same assumptions that we should be able to say that $BRF = BRDF \times \pi$. (We are already making a cosine correction for angle of source incidence.) Of course, this makes use of the Lambertian assumption which may not work well to correct for more specular surfaces.
- The BRDF values we are measuring for your samples are overall very low, and our uncertainties theoretically imply we don't think we can measure BRDF better than 1% absolute. However, some of the angular trends from your sample seem to be real and are not seen when we measure the same angles for our Spectralon reference diffuser.

Bidirectional reflectance factor (BRF) is the ratio of radiant flux [units W] reflected from a sample surface to the radiant flux which would be reflected from a Lambertian surface in the same configuration. A Lambertian surface reflects or emits light such that it has a constant radiance in all directions. Radiance [units W/m^2sr] is the radiant flux per projected area per steradian (solid angle). When a radiometer looks at a Lambertian disk, it does not directly measure the radiance at the radiometer. The voltage response of the radiometer to incident light is an integration over solid angle (defined by the aperture of the radiometer) and projected area (defined by the size of the cross-section of the radiometer FOV at the surface). Integrating radiance over projected area and solid angle gives the radiant flux. Provided the aperture, FOV, and distance to the Lambertian surface remain fixed, the radiometer should measure a constant voltage for all viewing configurations, since the radiance is constant in all directions.

This constant voltage is what allows RSG to use NIST BRF data of a Spectralon panel measured only in the 0/45 configuration (0 degree source zenith, 45 degree view zenith) to calculate the BRF of samples in arbitrary configurations. Consider the equation below which gives the definition of BRF:

$$BRF_{NIST} = \frac{\Phi_{Spectralon}}{\Phi_{Lambertian}}$$

The radiant flux is what is being measured by the radiometer, so (assuming a linear detector response) the equation can also be written as:

$$BRF_{NIST} = \frac{V_{Spectralon(0/45)}}{V_{Lambertian(0/45)}}$$

where V is the radiometer measurement of the surfaces. NIST provides BRF_{NIST} and RSG can measure $V_{Spectralon}$ in the Black Lab, so one can easily solve for the radiometer's response to a Lambertian reflector:

$$V_{Lambertian} = \frac{\Phi_{Spectralon}}{BRF_{NIST}}$$

Since the radiance is constant in all directions, this voltage value can be used to find the BRF of arbitrary samples in arbitrary viewing configurations.

One last consideration – when the surface is tilted relative to the source, the irradiance [W/m^2] at the surface decreases. The amount of light leaving the source is not changing, but the projected area of the surface decreases from the point of view of the source according to the cosine law. This similarly scales the voltage reading expected for a Lambertian reflector:

$$V_{Lambertian} = \frac{V_{Spectralon}}{BRF_{NIST}} \cos(\theta)$$

Putting this all together gives the equation for BRF of a sample for an arbitrary viewing/illumination configuration:

$$BRF_{Sample} = \frac{BRF_{NIST} V_{Sample}}{V_{Spectralon(0/45)} \cos(\theta)}$$

where $V_{Spectralon(0/45)}$ is the average voltage measured with the source and radiometer in the 0/45 configuration and theta is the source zenith angle from the sample measurement configuration.

Best,

Nik

From: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>

Sent: Thursday, June 25, 2026 11:07 PM

To: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>; Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>

Cc: Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>

Subject: Re: [BRDF] Debugging initial measurements

Hello Jacob

In the data plot, the + angle is toward the source.
In your hand drawing, the - angle toward the source.
Which one is correct?

And detector (receiver) aperture size?

Best
Heejoo

From: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>

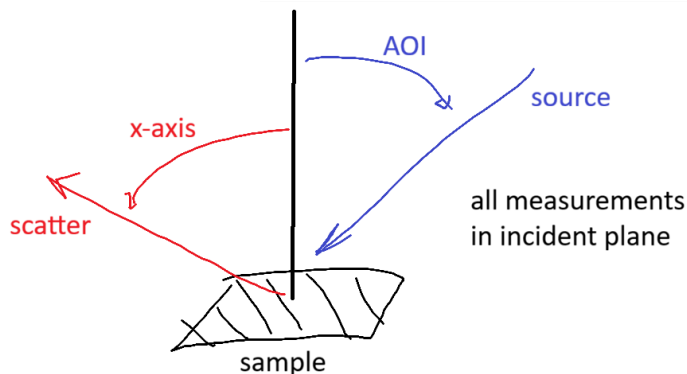
Sent: Tuesday, June 23, 2026 3:06 PM

To: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>

Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>

Subject: Re: [BRDF] Debugging initial measurements

Hello Heejoo,
Here is the coordinates of what is plotted:



Hello Nik,
I'll reach out a bit later today / tomorrow.

Best

-Jacob

From: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>

Sent: Tuesday, June 23, 2026 10:33 AM

To: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>

Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>

Subject: RE: [BRDF] Debugging initial measurements

Jacob,

Sorry for the delay, just getting back into the swing of things after our field work. Drop me a line this afternoon if you have some free time, we can catch up on some things I noticed in the reference data that will be helpful for helping us define which angles will and won't work. But long story short, we'll have to be careful to flag angles that not only have self shading issues, but there are also angles where the radiometer may see the source directly, and angles where the radiometer view is blocked by a stage component.

Thanks,

Nik

From: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>

Sent: Saturday, June 20, 2026 1:35 PM

To: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>

Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>

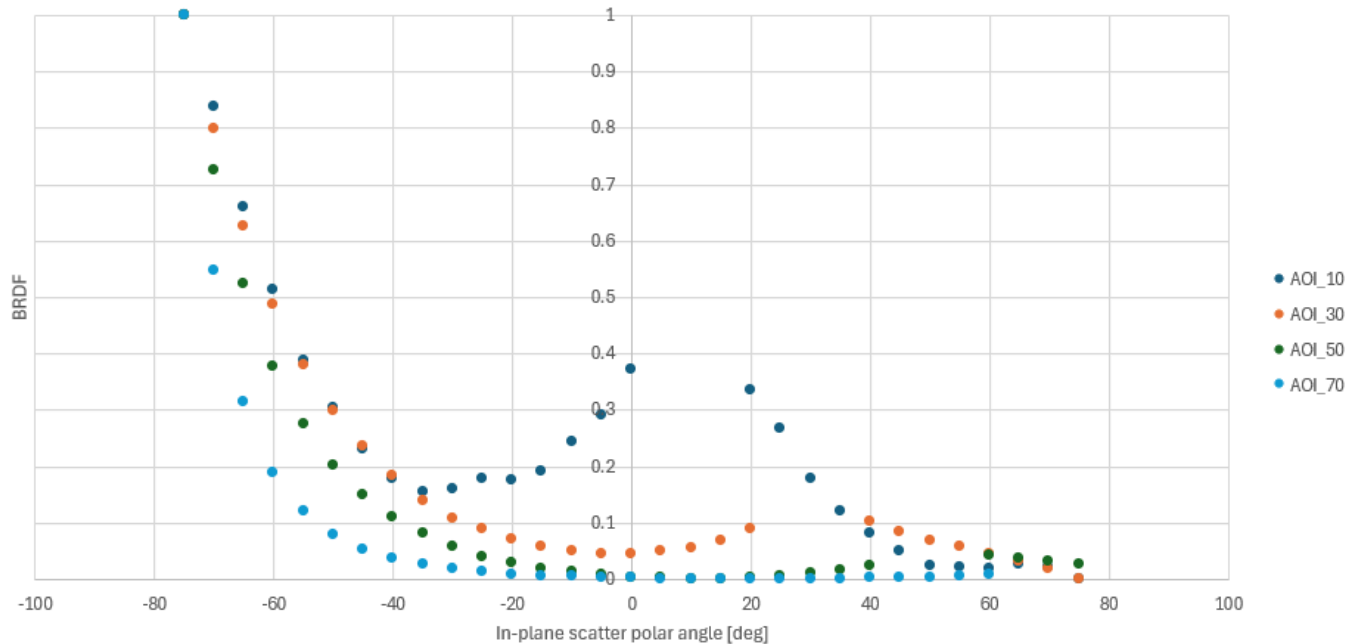
Subject: [BRDF] Debugging initial measurements

Hello Nik,

Just to keep in the loop, here is some of what I am finding with the in-plane Aeroglaze data so far:

- The angles which I thought were self-shading (and removed from the angle set) appear to be where the specular peak is see AOI_10 for example, where the angles (5, 10, 15) were not measured
- The behavior near -90deg is maybe due to stray light ----- not sure what we can do to mitigate this

[NORMALIZED] Blacklab validation of in-plane Aeroglaze at 631nm (2026/06/08)



Best

-Jacob